3.4 Bivariate Distributions

We generalize the concept of distribution of a random variable to the joint distribution of two random variables. In doing so, we introduce the joint p.f. for two discrete random variables, the joint p.d.f. for two continuous random variables, and the joint c.d.f. for any two random variables. We also introduce a joint hybrid of p.f. and p.d.f. for the case of one discrete random variable and one continuous random variable.

Example 3.4.1

Demands for Utilities. In Example 3.1.5, we found the distribution of the random variable X that represented the demand for water. But there is another random variable, Y, the demand for electricity, that is also of interest. When discussing two random variables at once, it is often convenient to put them together into an ordered pair, (X, Y). As early as Example 1.5.4 on page 19, we actually calculated some probabilities associated with the pair (X, Y). In that example, we defined two events A and B that we now can express as $A = \{X \geq 115\}$ and $B = \{Y \geq 110\}$. In Example 1.5.4, we computed $\Pr(A \cap B)$ and $\Pr(A \cup B)$. We can express $A \cap B$ and $A \cup B$ as events involving the pair (X, Y). For example, define the set of ordered pairs $C = \{(x, y) : x \geq 115 \text{ and } y \geq 110\}$ so that that the event $\{(X, Y) \in C)\} = A \cap B$. That is, the event that the pair of random variables lies in the set C is the same as the intersection of the two events A and B. In Example 1.5.4, we computed $\Pr(A \cap B) = 0.1198$. So, we can now assert that $\Pr((X, Y) \in C) = 0.1198$. ◀

Definition 3.4.1

Joint/Bivariate Distribution. Let X and Y be random variables. The *joint distribution* or *bivariate distribution* of X and Y is the collection of all probabilities of the form $\Pr[(X, Y) \in C]$ for all sets C of pairs of real numbers such that $\{(X, Y) \in C\}$ is an event.

It is a straightforward consequence of the definition of the joint distribution of X and Y that this joint distribution is itself a probability measure on the set of ordered pairs of real numbers. The set $\{(X, Y) \in C\}$ will be an event for every set C of pairs of real numbers that most readers will be able to imagine.

In this section and the next two sections, we shall discuss convenient ways to characterize and do computations with bivariate distributions. In Sec. 3.7, these considerations will be extended to the joint distribution of an arbitrary finite number of random variables.

Discrete Joint Distributions

Example 3.4.2

Theater Patrons. Suppose that a sample of 10 people is selected at random from a theater with 200 patrons. One random variable of interest might be the number X of people in the sample who are over 60 years of age, and another random variable might be the number Y of people in the sample who live more than 25 miles from the theater. For each ordered pair (x, y) with $x = 0, \ldots, 10$ and $y = 0, \ldots, 10$, we might wish to compute $\Pr((X, Y) = (x, y))$, the probability that there are x people in the sample who are over 60 years of age and there are y people in the sample who live more than 25 miles away. ◀

Definition 3.4.2

Discrete Joint Distribution. Let X and Y be random variables, and consider the ordered pair (X, Y). If there are only finitely or at most countably many different possible values (x, y) for the pair (X, Y), then we say that X and Y have a *discrete joint distribution*.

The two random variables in Example 3.4.2 have a discrete joint distribution.

Theorem 3.4.1 Suppose that two random variables X and Y each have a discrete distribution. Then X and Y have a discrete joint distribution.

Proof If both X and Y have only finitely many possible values, then there will be only a finite number of different possible values (x, y) for the pair (X, Y). On the other hand, if either X or Y or both can take a countably infinite number of possible values, then there will also be a countably infinite number of possible values for the pair (X, Y). In all of these cases, the pair (X, Y) has a discrete joint distribution. ■

When we define continuous joint distribution shortly, we shall see that the obvious analog of Theorem 3.4.1 is not true.

Definition 3.4.3 Joint Probability Function, p.f. The *joint probability function*, or the *joint p.f.*, of X and Y is defined as the function f such that for every point (x, y) in the xy-plane,

$$f(x, y) = \Pr(X = x \text{ and } Y = y).$$

The following result is easy to prove because there are at most countably many pairs (x, y) that must account for all of the probability a discrete joint distribution.

Theorem 3.4.2 Let X and Y have a discrete joint distribution. If (x, y) is not one of the possible values of the pair (X, Y), then $f(x, y) = 0$. Also,

$$\sum_{\text{All } (x,y)} f(x, y) = 1.$$

Finally, for each set C of ordered pairs,

$$\Pr[(X, Y) \in C] = \sum_{(x,y)\in C} f(x, y). \quad ■$$

Example 3.4.3 Specifying a Discrete Joint Distribution by a Table of Probabilities. In a certain suburban area, each household reported the number of cars and the number of television sets that they owned. Let X stand for the number of cars owned by a randomly selected household in this area. Let Y stand for the number of television sets owned by that same randomly selected household. In this case, X takes only the values 1, 2, and 3; Y takes only the values 1, 2, 3, and 4; and the joint p.f. f of X and Y is as specified in Table 3.2.

Table 3.2 Joint p.f. $f(x, y)$ for Example 3.4.3

	y			
x	1	2	3	4
1	0.1	0	0.1	0
2	0.3	0	0.1	0.2
3	0	0.2	0	0

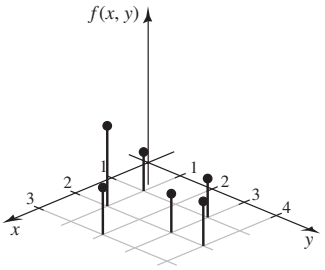


Figure 3.10 The joint p.f. of X and Y in Example 3.4.3.

This joint p.f. is sketched in Fig. 3.10. We shall determine the probability that the randomly selected household owns at least two of both cars and televisions. In symbols, this is $\Pr(X \geq 2 \text{ and } Y \geq 2)$.

By summing $f(x, y)$ over all values of $x \geq 2$ and $y \geq 2$, we obtain the value

$$\begin{aligned}\Pr(X \geq 2 \text{ and } Y \geq 2) &= f(2, 2) + f(2, 3) + f(2, 4) + f(3, 2) \\ &\quad + f(3, 3) + f(3, 4) \\ &= 0.5.\end{aligned}$$

Next, we shall determine the probability that the randomly selected household owns exactly one car, namely $\Pr(X = 1)$. By summing the probabilities in the first row of the table, we obtain the value

$$\Pr(X = 1) = \sum_{y=1}^{4} f(1, y) = 0.2. \quad \blacktriangleleft$$

Continuous Joint Distributions

Example 3.4.4 Demands for Utilities. Consider again the joint distribution of X and Y in Example 3.4.1. When we first calculated probabilities for these two random variables back in Example 1.5.4 on page 19 (even before we named them or called them random variables), we assumed that the probability of each subset of the sample space was proportional to the area of the subset. Since the area of the sample space is 29,204, the probability that the pair (X, Y) lies in a region C is the area of C divided by 29,204. We can also write this relation as

$$\Pr((X, Y) \in C\} = \int_C \int \frac{1}{29{,}204}\, dx\, dy, \tag{3.4.1}$$

assuming that the integral exists. $\blacktriangleleft$

If one looks carefully at Eq. (3.4.1), one will notice the similarity to Eqs. (3.2.2) and (3.2.1). We formalize this connection as follows.

Definition 3.4.4 Continuous Joint Distribution/Joint p.d.f./Support. Two random variables X and Y have a *continuous joint distribution* if there exists a nonnegative function f defined over the entire xy-plane such that for every subset C of the plane,

$$\Pr[(X, Y) \in C] = \int_C \int f(x, y)\, dx\, dy,$$

if the integral exists. The function f is called the *joint probability density function* (abbreviated *joint p.d.f.*) of X and Y. The closure of the set $\{(x, y) : f(x, y) > 0\}$ is called the *support of (the distribution of)* (X, Y).

Example 3.4.5

Demands for Utilities. In Example 3.4.4, it is clear from Eq. (3.4.1) that the joint p.d.f. of X and Y is the function

$$f(x, y) = \begin{cases} \dfrac{1}{29{,}204} & \text{for } 4 \le x \le 200 \text{ and } 1 \le y \le 150, \\ 0 & \text{otherwise.} \end{cases} \tag{3.4.2}$$

◀

It is clear from Definition 3.4.4 that the joint p.d.f. of two random variables characterizes their joint distribution. The following result is also straightforward.

Theorem 3.4.3

A joint p.d.f. must satisfy the following two conditions:

$$f(x, y) \ge 0 \quad \text{for } -\infty < x < \infty \text{ and } -\infty < y < \infty,$$

and

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1.$$

■

Any function that satisfies the two displayed formulas in Theorem 3.4.3 is the joint p.d.f. for some probability distribution.

An example of the graph of a joint p.d.f. is presented in Fig. 3.11. The total volume beneath the surface $z = f(x, y)$ and above the xy-plane must be 1. The probability that the pair (X, Y) will belong to the rectangle C is equal to the volume of the solid figure with base A shown in Fig. 3.11. The top of this solid figure is formed by the surface $z = f(x, y)$.

In Sec. 3.5, we will show that if X and Y have a continuous joint distribution, then X and Y each have a continuous distribution when considered separately. This seems reasonable intutively. However, the converse of this statement is not true, and the following result helps to show why.

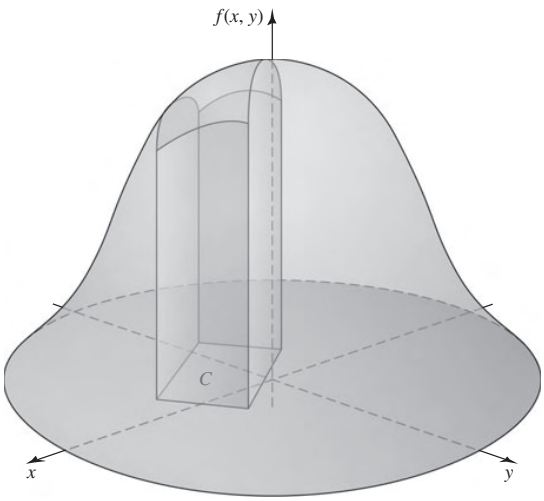


Figure 3.11 An example of a joint p.d.f.

Theorem 3.4.4 For every continuous joint distribution on the xy-plane, the following two statements hold:

i. Every individual point, and every infinite sequence of points, in the xy-plane has probability 0.

ii. Let f be a continuous function of one real variable defined on a (possibly unbounded) interval (a, b). The sets $\{(x, y) : y = f(x), a < x < b\}$ and $\{(x, y) : x = f(y), a < y < b\}$ have probability 0.

Proof According to Definition 3.4.4, the probability that a continuous joint distribution assigns to a specified region of the xy-plane can be found by integrating the joint p.d.f. $f(x, y)$ over that region, if the integral exists. If the region is a single point, the integral will be 0. By Axiom 3 of probability, the probability for any countable collection of points must also be 0. The integral of a function of two variables over the graph of a continuous function in the xy-plane is also 0. ■

Example 3.4.6 Not a Continuous Joint Distribution. It follows from (ii) of Theorem 3.4.4 that the probability that (X, Y) will lie on each specified straight line in the plane is 0. If X has a continuous distribution and if $Y = X$, then both X and Y have continuous distributions, but the probability is 1 that (X, Y) lies on the straight line $y = x$. Hence, X and Y cannot have a continuous joint distribution. ◀

Example 3.4.7 Calculating a Normalizing Constant. Suppose that the joint p.d.f. of X and Y is specified as follows:

$$f(x, y) = \begin{cases} cx^2y & \text{for } x^2 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

We shall determine the value of the constant c.

The support S of (X, Y) is sketched in Fig. 3.12. Since $f(x, y) = 0$ outside S, it follows that

$$\begin{aligned} \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x, y)\,dx\,dy &= \int_S\int f(x, y)\,dx\,dy \\ &= \int_{-1}^{1}\int_{x^2}^{1} cx^2y\,dy\,dx = \frac{4}{21}c. \end{aligned} \tag{3.4.3}$$

Since the value of this integral must be 1, the value of c must be 21/4.

The limits of integration on the last integral in (3.4.3) were determined as follows. We have our choice of whether to integrate x or y as the inner integral, and we chose y. So, we must find, for each x, the interval of y values over which to integrate. From Fig. 3.12, we see that, for each x, y runs from the curve where $y = x^2$ to the line where $y = 1$. The interval of x values for the outer integral is from -1 to 1 according to Fig. 3.12. If we had chosen to integrate x on the inside, then for each y, we see that x runs from $-\sqrt{y}$ to $\sqrt{y}$, while y runs from 0 to 1. The final answer would have been the same. ◀

Example 3.4.8 Calculating Probabilities from a Joint p.d.f. For the joint distribution in Example 3.4.7, we shall now determine the value of $\Pr(X \ge Y)$.

The subset S_0 of S where $x \ge y$ is sketched in Fig. 3.13. Hence,

$$\Pr(X \ge Y) = \int_{S_0}\int f(x, y)\,dx\,dy = \int_0^1\int_{x^2}^{x} \frac{21}{4}x^2y\,dy\,dx = \frac{3}{20}. \quad ◀$$

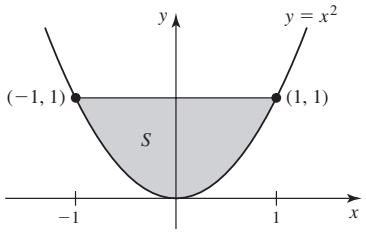


Figure 3.12 The support S of (X, Y) in Example 3.4.8.

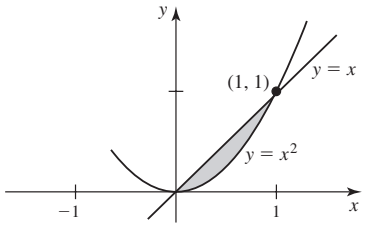


Figure 3.13 The subset S_0 of the support S where $x \geq y$ in Example 3.4.8.

Example 3.4.9

Determining a Joint p.d.f. by Geometric Methods. Suppose that a point (X, Y) is selected at random from inside the circle $x^2 + y^2 \leq 9$. We shall determine the joint p.d.f. of X and Y.

The support of (X, Y) is the set S of points on and inside the circle $x^2 + y^2 \leq 9$. The statement that the point (X, Y) is selected at random from inside the circle is interpreted to mean that the joint p.d.f. of X and Y is constant over S and is 0 outside S. Thus,

$$f(x, y) = \begin{cases} c & \text{for } (x, y) \in S, \\ 0 & \text{otherwise.} \end{cases}$$

We must have

$$\int_S \int f(x, y)\, dx\, dy = c \times (\text{area of } S) = 1.$$

Since the area of the circle S is 9π, the value of the constant c must be $1/(9\pi)$. ◀

Mixed Bivariate Distributions

Example 3.4.10

A Clinical Trial. Consider a clinical trial (such as the one described in Example 2.1.12) in which each patient with depression receives a treatment and is followed to see whether they have a relapse into depression. Let X be the indicator of whether or not the first patient is a "success" (no relapse). That is $X = 1$ if the patient does not relapse and $X = 0$ if the patient relapses. Also, let P be the proportion of patients who have no replapse among all patients who might receive the treatment. It is clear that X must have a discrete distribution, but it might be sensible to think of P as a continuous random variable taking its value anywhere in the interval $[0, 1]$. Even though X and P can have neither a joint discrete distribution nor a joint continuous distribution, we can still be interested in the joint distribution of X and P. ◀

Prior to Example 3.4.10, we have discussed bivariate distributions that were either discrete or continuous. Occasionally, one must consider a mixed bivariate distribution in which one of the random variables is discrete and the other is continuous. We shall use a function $f(x, y)$ to characterize such a joint distribution in much the same way that we use a joint p.f. to characterize a discrete joint distribution or a joint p.d.f. to characterize a continuous joint distribution.

Definition 3.4.5 Joint p.f./p.d.f. Let X and Y be random variables such that X is discrete and Y is continuous. Suppose that there is a function $f(x, y)$ defined on the xy-plane such that, for every pair A and B of subsets of the real numbers,

$$\Pr(X \in A \text{ and } Y \in B) = \int_B \sum_{x \in A} f(x, y)dy, \tag{3.4.4}$$

if the integral exists. Then the function f is called the *joint p.f./p.d.f.* of X and Y.

Clearly, Definition 3.4.5 can be modified in an obvious way if Y is discrete and X is continuous. Every joint p.f./p.d.f. must satisfy two conditions. If X is the discrete random variable with possible values $x_1, x_2, \ldots$ and Y is the continuous random variable, then $f(x, y) \geq 0$ for all x, y and

$$\int_{-\infty}^{\infty} \sum_{i=1}^{\infty} f(x_i, y)dy = 1. \tag{3.4.5}$$

Because f is nonnegative, the sum and integral in Eqs. (3.4.4) and (3.4.5) can be done in whichever order is more convenient.

Note: Probabilities of More General Sets. For a general set C of pairs of real numbers, we can compute $\Pr((X, Y) \in C)$ using the joint p.f./p.d.f. of X and Y. For each x, let $C_x = \{y : (x, y) \in C\}$. Then

$$\Pr((X, Y) \in C) = \sum_{\text{All } x} \int_{C_x} f(x, y)dy,$$

if all of the integrals exist. Alternatively, for each y, define $C^y = \{x : (x, y) \in C\}$, and then

$$\Pr((X, Y) \in C) = \int_{-\infty}^{\infty} \left[\sum_{x \in C^y} f(x, y) \right] dy,$$

if the integral exists.

Example 3.4.11 A Joint p.f./p.d.f. Suppose that the joint p.f./p.d.f. of X and Y is

$$f(x, y) = \frac{xy^{x-1}}{3}, \quad \text{for } x = 1, 2, 3 \text{ and } 0 < y < 1.$$

We should check to make sure that this function satisfies (3.4.5). It is easier to integrate over the y values first, so we compute

$$\sum_{x=1}^{3} \int_0^1 \frac{xy^{x-1}}{3} dy = \sum_{x=1}^{3} \frac{1}{3} = 1.$$

Suppose that we wish to compute the probability that $Y \geq 1/2$ and $X \geq 2$. That is, we want $\Pr(X \in A \text{ and } Y \in B)$ with $A = [2, \infty)$ and $B = [1/2, \infty)$. So, we apply Eq. (3.4.4)

to get the probability

$$\sum_{x=2}^{3}\int_{1/2}^{1}\frac{xy^{x-1}}{3}dy=\sum_{x=2}^{3}\left(\frac{1-(1/2)^x}{3}\right)=0.5417.$$

For illustration, we shall compute the sum and integral in the other order also. For each $y\in[1/2,1)$, $\sum_{x=2}^{3}f(x,y)=2y/3+y^2$. For $y\geq 1/2$, the sum is 0. So, the probability is

$$\int_{1/2}^{1}\left[\frac{2}{3}y+y^2\right]dy=\frac{1}{3}\left[1-\left(\frac{1}{2}\right)^2\right]+\frac{1}{3}\left[1-\left(\frac{1}{2}\right)^3\right]=0.5417.$$ ◀

Example 3.4.12

A Clinical Trial. A possible joint p.f./p.d.f. for X and P in Example 3.4.10 is

$$f(x,p)=p^x(1-p)^{1-x},\quad \text{for } x=0,1 \text{ and } 0<p<1.$$

Here, X is discrete and P is continuous. The function f is nonnegative, and the reader should be able to demonstrate that it satisfies (3.4.5). Suppose that we wish to compute $\Pr(X\leq 0 \text{ and } P\leq 1/2)$. This can be computed as

$$\int_{0}^{1/2}(1-p)dp=-\frac{1}{2}[(1-1/2)^2-(1-0)^2]=\frac{3}{8}.$$

Suppose that we also wish to compute $\Pr(X=1)$. This time, we apply Eq. (3.4.4) with $A=\{1\}$ and $B=(0,1)$. In this case,

$$\Pr(X=1)=\int_{0}^{1}p\,dp=\frac{1}{2}.$$ ◀

A more complicated type of joint distribution can also arise in a practical problem.

Example 3.4.13

A Complicated Joint Distribution. Suppose that X and Y are the times at which two specific components in an electronic system fail. There might be a certain probability p $(0<p<1)$ that the two components will fail at the same time and a certain probability $1-p$ that they will fail at different times. Furthermore, if they fail at the same time, then their common failure time might be distributed according to a certain p.d.f. $f(x)$; if they fail at different times, then these times might be distributed according to a certain joint p.d.f. $g(x,y)$.

The joint distribution of X and Y in this example is not continuous, because there is positive probability p that (X,Y) will lie on the line $x=y$. Nor does the joint distribution have a joint p.f./p.d.f. or any other simple function to describe it. There are ways to deal with such joint distributions, but we shall not discuss them in this text. ◀

Bivariate Cumulative Distribution Functions

The first calculation in Example 3.4.12, namely, $\Pr(X\leq 0 \text{ and } Y\leq 1/2)$, is a generalization of the calculation of a c.d.f. to a bivariate distribution. We formalize the generalization as follows.

Definition 3.4.6

Joint (Cumulative) Distribution Function/c.d.f. The *joint distribution function* or *joint cumulative distribution function* (*joint c.d.f.*) of two random variables X and Y is

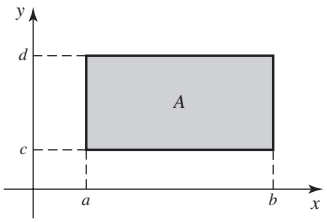


Figure 3.14 The probability of a rectangle.

defined as the function F such that for all values of x and y ($-\infty < x < \infty$ and $-\infty < y < \infty$),

$$F(x, y) = \Pr(X \le x \text{ and } Y \le y).$$

It is clear from Definition 3.4.6 that $F(x, y)$ is monotone increasing in x for each fixed y and is monotone increasing in y for each fixed x.

If the joint c.d.f. of two arbitrary random variables X and Y is F, then the probability that the pair (X, Y) will lie in a specified rectangle in the xy-plane can be found from F as follows: For given numbers $a < b$ and $c < d$,

$$\begin{aligned}
&\Pr(a < X \le b \text{ and } c < Y \le d)\\
&\quad = \Pr(a < X \le b \text{ and } Y \le d) - \Pr(a < X \le b \text{ and } Y \le c)\\
&\quad = [\Pr(X \le b \text{ and } Y \le d) - \Pr(X \le a \text{ and } Y \le d)]\\
&\qquad\quad -[\Pr(X \le b \text{ and } Y \le c) - \Pr(X \le a \text{ and } Y \le c)]\\
&\quad = F(b, d) - F(a, d) - F(b, c) + F(a, c).
\end{aligned} \tag{3.4.6}$$

Hence, the probability of the rectangle C sketched in Fig. 3.14 is given by the combination of values of F just derived. It should be noted that two sides of the rectangle are included in the set C and the other two sides are excluded. Thus, if there are points or line segments on the boundary of C that have positive probability, it is important to distinguish between the weak inequalities and the strict inequalities in Eq. (3.4.6).

Theorem 3.4.5 Let X and Y have a joint c.d.f. F. The c.d.f. F_1 of just the single random variable X can be derived from the joint c.d.f. F as $F_1(x) = \lim_{y\to\infty} F(x, y)$. Similarly, the c.d.f. F_2 of Y equals $F_2(y) = \lim_{x\to\infty} F(x, y)$, for $0 < y < \infty$.

Proof We prove the claim about F_1 as the claim about F_2 is similar. Let $-\infty < x < \infty$. Define

$$\begin{aligned}
B_0 &= \{X \le x \text{ and } Y \le 0\},\\
B_n &= \{X \le x \text{ and } n - 1 < Y \le n\}, \quad \text{for } n = 1, 2, \ldots,\\
A_m &= \bigcup_{n=0}^{m} B_n, \quad \text{for } m = 1, 2, \ldots.
\end{aligned}$$

Then $\{X \le x\} = \bigcup_{n=-0}^{\infty} B_n$, and $A_m = \{X \le x \text{ and } Y \le m\}$ for $m = 1, 2, \ldots$. It follows that $\Pr(A_m) = F(x, m)$ for each m. Also,

$$F_1(x) = \Pr(X \le x) = \Pr\left(\bigcup_{n=1}^{\infty} B_n\right)$$
$$= \sum_{n=0}^{\infty} \Pr(B_n) = \lim_{m\to\infty} \Pr(A_m)$$
$$= \lim_{m\to\infty} F(x, m) = \lim_{y\to\infty} F(x, y),$$

where the third equality follows from countable additivity and the fact that the B_n events are disjoint, and the last equality follows from the fact that $F(x, y)$ is monotone increasing in y for each fixed x. ■

Other relationships involving the univariate distribution of X, the univariate distribution of Y, and their joint bivariate distribution will be presented in the next section.

Finally, if X and Y have a continuous joint distribution with joint p.d.f. f, then the joint c.d.f. at (x, y) is

$$F(x, y) = \int_{-\infty}^{y} \int_{-\infty}^{x} f(r, s)\, dr\, ds.$$

Here, the symbols r and s are used simply as dummy variables of integration. The joint p.d.f. can be derived from the joint c.d.f. by using the relations

$$f(x, y) = \frac{\partial^2 F(x, y)}{\partial x \partial y} = \frac{\partial^2 F(x, y)}{\partial y \partial x}$$

at every point (x, y) at which these second-order derivatives exist.

Example 3.4.14 Determining a Joint p.d.f. from a Joint c.d.f. Suppose that X and Y are random variables that take values only in the intervals $0 \le X \le 2$ and $0 \le Y \le 2$. Suppose also that the joint c.d.f. of X and Y, for $0 \le x \le 2$ and $0 \le y \le 2$, is as follows:

$$F(x, y) = \frac{1}{16} xy(x + y). \tag{3.4.7}$$

We shall first determine the c.d.f. F_1 of just the random variable X and then determine the joint p.d.f. f of X and Y.

The value of $F(x, y)$ at any point (x, y) in the xy-plane that does not represent a pair of possible values of X and Y can be calculated from (3.4.7) and the fact that $F(x, y) = \Pr(X \le x \text{ and } Y \le y)$. Thus, if either $x < 0$ or $y < 0$, then $F(x, y) = 0$. If both $x > 2$ and $y > 2$, then $F(x, y) = 1$. If $0 \le x \le 2$ and $y > 2$, then $F(x, y) = F(x, 2)$, and it follows from Eq. (3.4.7) that

$$F(x, y) = \frac{1}{8} x(x + 2).$$

Similarly, if $0 \le y \le 2$ and $x > 2$, then

$$F(x, y) = \frac{1}{8} y(y + 2).$$

The function $F(x, y)$ has now been specified for every point in the xy-plane.

By letting $y \to \infty$, we find that the c.d.f. of just the random variable X is

$$F_1(x) = \begin{cases} 0 & \text{for } x < 0, \\ \frac{1}{8} x(x + 2) & \text{for } 0 \le x \le 2, \\ 1 & \text{for } x > 2. \end{cases}$$

Furthermore, for $0 < x < 2$ and $0 < y < 2$,

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = \frac{1}{8}(x + y).$$

Also, if $x < 0$, $y < 0$, $x > 2$, or $y > 2$, then

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = 0.$$

Hence, the joint p.d.f. of X and Y is

$$f(x, y) = \begin{cases} \frac{1}{8}(x + y) & \text{for } 0 < x < 2 \text{ and } 0 < y < 2, \\ 0 & \text{otherwise.} \end{cases}$$ ◀

Example 3.4.15

Demands for Utilities. We can compute the joint c.d.f. for water and electric demand in Example 3.4.4 by using the joint p.d.f. that was given in Eq. (3.4.2). If either $x \le 4$ or $y \le 1$, then $F(x, y) = 0$ because either $X \le x$ or $Y \le y$ would be impossible. Similarly, if both $x \ge 200$ and $y \ge 150$, $F(x, y) = 1$ because both $X \le x$ and $Y \le y$ would be sure events. For other values of x and y, we compute

$$F(x, y) = \begin{cases} \displaystyle\int_4^x \int_1^y \frac{1}{29{,}204} dy dx = \frac{xy}{29{,}204} & \text{for } 4 \le x \le 200, 1 \le y \le 150, \\ \displaystyle\int_4^x \int_1^{150} \frac{1}{29{,}204} dy dx = \frac{x}{196} & \text{for } 4 \le x \le 200, y > 150, \\ \displaystyle\int_4^{200} \int_1^y \frac{1}{29{,}204} dy dx = \frac{y}{149} & \text{for } x > 200, 1 \le y \le 150. \end{cases}$$

The reason that we need three cases in the formula for $F(x, y)$ is that the joint p.d.f. in Eq. (3.4.2) drops to 0 when x crosses above 200 or when y crosses above 150; hence, we never want to integrate 1/29,204 beyond $x = 200$ or beyond $y = 150$. If one takes the limit as $y \to \infty$ of $F(x, y)$ (for fixed $4 \le x \le 200$), one gets the second case in the formula above, which then is the c.d.f. of X, $F_1(x)$. Similarly, if one takes the $\lim_{x \to \infty} F(x, y)$ (for fixed $1 \le y \le 150$), one gets the third case in the formula, which then is the c.d.f. of Y, $F_2(y)$. ◀

Summary

The joint c.d.f. of two random variables X and Y is $F(x, y) = \Pr(X \le x \text{ and } Y \le y)$. The joint p.d.f. of two continuous random variables is a nonnegative function f such that the probability of the pair (X, Y) being in a set C is the integral of $f(x, y)$ over the set C, if the integral exists. The joint p.d.f. is also the second mixed partial derivative of the joint c.d.f. with respect to both variables. The joint p.f. of two discrete random variables is a nonnegative function f such that the probability of the pair (X, Y) being in a set C is the sum of $f(x, y)$ over all points in C. A joint p.f. can be strictly positive at countably many pairs (x, y) at most. The joint p.f./p.d.f. of a discrete random variable X and a continuous random variable Y is a nonnegative function f such that the probability of the pair (X, Y) being in a set C is obtained by summing $f(x, y)$ over all x such that $(x, y) \in C$ for each y and then integrating the resulting function of y.

Exercises

1. Suppose that the joint p.d.f. of a pair of random variables (X, Y) is constant on the rectangle where $0 \le x \le 2$ and $0 \le y \le 1$, and suppose that the p.d.f. is 0 off of this rectangle.

a. Find the constant value of the p.d.f. on the rectangle.

b. Find $\Pr(X \ge Y)$.

2. Suppose that in an electric display sign there are three light bulbs in the first row and four light bulbs in the second row. Let X denote the number of bulbs in the first row that will be burned out at a specified time t, and let Y denote the number of bulbs in the second row that will be burned out at the same time t. Suppose that the joint p.f. of X and Y is as specified in the following table:

	Y				
X	0	1	2	3	4
0	0.08	0.07	0.06	0.01	0.01
1	0.06	0.10	0.12	0.05	0.02
2	0.05	0.06	0.09	0.04	0.03
3	0.02	0.03	0.03	0.03	0.04

Determine each of the following probabilities:

a. $\Pr(X = 2)$ **b.** $\Pr(Y \ge 2)$

c. $\Pr(X \le 2 \text{ and } Y \le 2)$ **d.** $\Pr(X = Y)$

e. $\Pr(X > Y)$

3. Suppose that X and Y have a discrete joint distribution for which the joint p.f. is defined as follows:

$$f(x, y) = \begin{cases} c|x + y| & \text{for } x = -2, -1, 0, 1, 2 \text{ and} \\ & \quad y = -2, -1, 0, 1, 2, \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the value of the constant c; **(b)** $\Pr(X = 0 \text{ and } Y = -2)$; **(c)** $\Pr(X = 1)$; **(d)** $\Pr(|X - Y| \le 1)$.

4. Suppose that X and Y have a continuous joint distribution for which the joint p.d.f. is defined as follows:

$$f(x, y) = \begin{cases} cy^2 & \text{for } 0 \le x \le 2 \text{ and } 0 \le y \le 1, \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the value of the constant c; **(b)** $\Pr(X + Y > 2)$; **(c)** $\Pr(Y < 1/2)$; **(d)** $\Pr(X \le 1)$; **(e)** $\Pr(X = 3Y)$.

5. Suppose that the joint p.d.f. of two random variables X and Y is as follows:

$$f(x, y) = \begin{cases} c(x^2 + y) & \text{for } 0 \le y \le 1 - x^2, \\ 0 & \text{otherwise.} \end{cases}$$

Determine **(a)** the value of the constant c; **(b)** $\Pr(0 \le X \le 1/2)$; **(c)** $\Pr(Y \le X + 1)$; **(d)** $\Pr(Y = X^2)$.

6. Suppose that a point (X, Y) is chosen at random from the region S in the xy-plane containing all points (x, y) such that $x \ge 0$, $y \ge 0$, and $4y + x \le 4$.

a. Determine the joint p.d.f. of X and Y.

b. Suppose that S_0 is a subset of the region S having area α and determine $\Pr[(X, Y) \in S_0]$.

7. Suppose that a point (X, Y) is to be chosen from the square S in the xy-plane containing all points (x, y) such that $0 \le x \le 1$ and $0 \le y \le 1$. Suppose that the probability that the chosen point will be the corner (0, 0) is 0.1, the probability that it will be the corner (1, 0) is 0.2, the probability that it will be the corner (0, 1) is 0.4, and the probability that it will be the corner (1, 1) is 0.1. Suppose also that if the chosen point is not one of the four corners of the square, then it will be an interior point of the square and will be chosen according to a constant p.d.f. over the interior of the square. Determine **(a)** $\Pr(X \le 1/4)$ and **(b)** $\Pr(X + Y \le 1)$.

8. Suppose that X and Y are random variables such that (X, Y) must belong to the rectangle in the xy-plane containing all points (x, y) for which $0 \le x \le 3$ and $0 \le y \le 4$. Suppose also that the joint c.d.f. of X and Y at every point (x, y) in this rectangle is specified as follows:

$$F(x, y) = \frac{1}{156} xy(x^2 + y).$$

Determine **(a)** $\Pr(1 \le X \le 2 \text{ and } 1 \le Y \le 2)$; **(b)** $\Pr(2 \le X \le 4 \text{ and } 2 \le Y \le 4)$; **(c)** the c.d.f. of Y; **(d)** the joint p.d.f. of X and Y; **(e)** $\Pr(Y \le X)$.

9. In Example 3.4.5, compute the probability that water demand X is greater than electric demand Y.

10. Let Y be the rate (calls per hour) at which calls arrive at a switchboard. Let X be the number of calls during a two-hour period. A popular choice of joint p.f./p.d.f. for (X, Y) in this example would be one like

$$f(x, y) = \begin{cases} \frac{(2y)^x}{x!} e^{-3y} & \text{if } y > 0 \text{ and } x = 0, 1, \ldots, \\ 0 & \text{otherwise.} \end{cases}$$

a. Verify that f is a joint p.f./p.d.f. *Hint:* First, sum over the x values using the well-known formula for the power series expansion of e^{2y}.

b. Find $\Pr(X = 0)$.

11. Consider the clinical trial of depression drugs in Example 2.1.4. Suppose that a patient is selected at random from the 150 patients in that study and we record Y, an

Table 3.3 Proportions in clinical depression study for Exercise 11

	Treatment group (Y)			
Response (X)	Imipramine (1)	Lithium (2)	Combination (3)	Placebo (4)
Relapse (0)	0.120	0.087	0.146	0.160
No relapse (1)	0.147	0.166	0.107	0.067

indicator of the treatment group for that patient, and X, an indicator of whether or not the patient relapsed. Table 3.3 contains the joint p.f. of X and Y.

a. Calculate the probability that a patient selected at random from this study used Lithium (either alone or in combination with Imipramine) and did not relapse.

b. Calculate the probability that the patient had a relapse (without regard to the treatment group).

3.5 Marginal Distributions

Earlier in this chapter, we introduced distributions for random variables, and in Sec. 3.4 we discussed a generalization to joint distributions of two random variables simultaneously. Often, we start with a joint distribution of two random variables and we then want to find the distribution of just one of them. The distribution of one random variable X computed from a joint distribution is also called the marginal distribution of X. Each random variable will have a marginal c.d.f. as well as a marginal p.d.f. or p.f. We also introduce the concept of independent random variables, which is a natural generalization of independent events.

Deriving a Marginal p.f. or a Marginal p.d.f.

We have seen in Theorem 3.4.5 that if the joint c.d.f. F of two random variables X and Y is known, then the c.d.f. F_1 of the random variable X can be derived from F. We saw an example of this derivation in Example 3.4.15. If X has a continuous distribution, we can also derive the p.d.f. of X from the joint distribution.

Example 3.5.1

Demands for Utilities. Look carefully at the formula for $F(x, y)$ in Example 3.4.15, specifically the last two branches that we identified as $F_1(x)$ and $F_2(y)$, the c.d.f.'s of the two individual random variables X and Y. It is apparent from those two formulas and Theorem 3.3.5 that the p.d.f. of X alone is

$$f_1(x) = \begin{cases} \frac{1}{196} & \text{for } 4 \le x \le 200, \\ 0 & \text{otherwise,} \end{cases}$$

which matches what we already found in Example 3.2.1. Similarly, the p.d.f. of Y alone is

$$f_2(y) = \begin{cases} \frac{1}{149} & \text{for } 1 \le y \le 150, \\ 0 & \text{otherwise.} \end{cases}$$

◀

The ideas employed in Example 3.5.1 lead to the following definition.